

Scalability & Future Plan

Date	05-10-2026
Team ID	6ab2256ffc67a5bc022256a7
Project Name	Script_ Controlled ACL _ Restrict Record Access Based on Field Value
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Synchronous Script Execution	Complex GlideRecord calls in ACL scripts increase transaction response time during high-concurrency list queries.	High
2	Hardcoded Role Checks	Static role evaluation inside scripts requires manual Update Set deployments when department roles change.	Medium
3	Single Instance Access Logging	Security audit logs are stored locally in ServiceNow sys_audit, creating high table storage bloat over time.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single department / team evaluation	Implement cached session variables (gs.getSession().getClientData) to prevent repeated database queries.
2	Data Storage	Standard table-level security rules	Archive historical audit logs automatically to external cloud storage to optimize system database performance.
3	Performance	Evaluates logic on every record fetch	Leverage indexed condition builder rules alongside lightweight script includes for sub-millisecond evaluation.
4	Security	Role-based field level security	Integrate Attribute-Based Access Control (ABAC) with dynamic user attributes from Identity Provider (IDP).

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Dynamic System Property Configuration	Q1 2027	Allows security admins to adjust field restriction rules via System Properties without code changes.
Phase 3	Automated Access Request Workflow	Q2 2027	Provides temporary elevated field access via Service Catalog requests with automated timeout.
Phase 4	SIEM Integration & Real-time Auditing	Q3 2027	Streams unauthorized access attempt logs directly to enterprise Splunk / Sentinel dashboards.